

Lecture-3

Modern Periodic Table

Book Ref.

Introduction to Modern Inorganic Chemistry

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What is a Periodic Table?

- ▶ A tabular arrangement of elements in rows and columns, highlighting the regular repetition of properties of the elements, is called a periodic table.

Periodic Table

1

1IA

1A

2

IIA

2A

3

IIIB

3B

4

IVB

4B

5

VB

5B

6

VIB

6B

7

VII B

7B

8

VIII

8

9

VIII

8

10

VIII

8

11

IB

1B

12

IIB

2B

13

IIIA

3A

14

IVA

4A

15

VA

5A

16

VIA

6A

17

VIIA

7A

18

VIIIA

8A

1

H

1.01

Hydrogen

Atomic Number

Symbol

Relative Atomic Mass

Name

3

Li

Lithium

6.941

4

Be

Beryllium

9.012

11

Na

Sodium

22.99

12

Mg

Magnesium

24.305

19

K

Potassium

39.098

20

Ca

Calcium

40.078

27

Co

Cobalt

58.933

28

Ni

Nickel

58.693

37

Rb

Rubidium

85.468

38

Sr

Strontium

87.62

45

Rh

Rhodium

102.906

46

Pd

Palladium

106.42

55

Cs

Cesium

132.905

56

Ba

Barium

137.328

63

Eu

Europium

151.964

64

Gd

Gadolinium

157.25

71

Lu

Lutetium

174.967

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.948

81

Tl

Thallium

204.383

82

Pb

Lead

207.2

89

Ac

Actinium

227.028

90

Th

Thorium

232.038

91

Pa

Protactinium

231.036

92

U

Uranium

238.029

93

Np

Neptunium

237.048

94

Pu

Plutonium

244.064

95

Am

Americium

243.061

96

Cm

Curium

247.070

97

Bk

Berkelium

247.070

98

Cf

Californium

251.080

99

Es

Einsteinium

[254]

100

Fm

Fermium

257.095

101

Md

Mendelevium

258.1

102

No

Nobelium

259.101

103

Lr

Lawrencium

[262]

Alkali Metal

Alkaline Earth

Transition Metal

Basic Metal

Semimetal

Nonmetal

Halogen

Noble Gas

Lanthanide

Actinide

Law of Periodicity

- ▶ **Modern Statement of the Law of Periodicity (Moseley):**
‘The modern periodic law states that the properties of the elements repeat after certain regular intervals when these elements are arranged in order of their increasing atomic numbers’

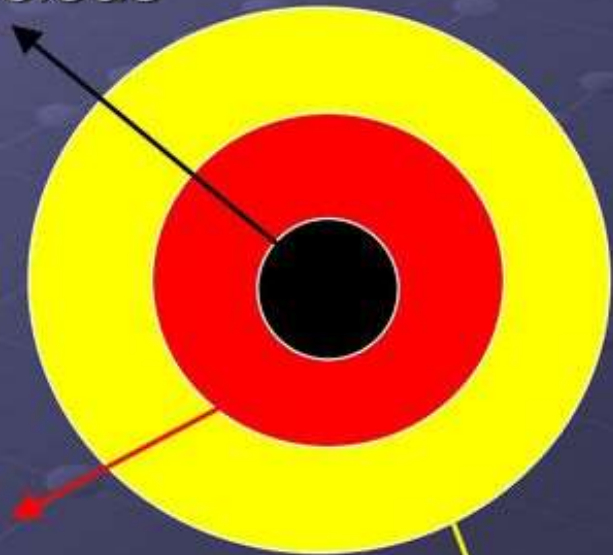
cause of periodicity

- Repetition of similar outer electronic configurations after certain regular intervals.
- Since the nucleus is situated at the center of an atom and is shielded by electrons in the outermost level, therefore, the atomic mass has little effect on the chemical properties of the elements.
- Electrons are exposed to the environment. Hence, they can interact with other atoms. As a result, the physical and chemical properties of the elements depend upon their atomic numbers rather than the atomic masses.

Shielding Effect

- The shielding effect is the reduction of attractive force between the nucleus (+) and its outer electrons (-) due to the blocking affect of the inner electrons

Nucleus



Shielding electrons

Valence electrons 'shielded' by inner electrons

Main Features of Modern Periodic Table

- ▶ Modern periodic table is based upon the electronic configuration of the elements. It consists of 18 vertical columns (Groups) and 7 horizontal rows (Periods).
- ▶ **Groups:** Elements that have similar chemical properties are grouped in columns called groups. All the members of a Group have the same valence configuration but different principal quantum numbers. These groups are numbered from 1 to 18.
- ▶ The groups are also divided into two sub- groups: A and B. These are IA(1), IIA(2), IIIB(3), IVB(4), VB(5), VIB(6), VIIB(7), VIIIB(8,9,10), IB(11), IIB(12), IIIA(13), IVA(14), VA(15), VIA(16), VIIA(17) and VIIIA (0,18).

V·T·E	Groups in the periodic table																		[hide]
IUPAC group	1	2	3 ^a	^a	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Mendeleev (I–VIII)	I	II	III		IV	V	VI	VII	VIII			I	II	III	IV	V	VI	VII	^b
CAS (US, A-B-A)	IA	IIA	IIIB		IVB	VB	VIB	VII B	VIII B			IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
old IUPAC (Europe, A-B)	IA	IIA	IIIA		IVA	VA	VIA	VIIA	VIII			IB	IIB	IIIB	IVB	VB	VIB	VIIB	0
Trivial name	Alkali metals	Alkaline earth metals ^r										Coinage metals	Volatile metals	Icosagens	Crystallogens	Pnictogens ^r	Chalcogens ^r	Halogens ^r	Noble gases ^r
Name by element ^f	Lithium group	Beryllium group	Scandium group		Titanium group	Vanadium group	Chromium group	Manganese group	Iron group	Cobalt group	Nickel group	Copper group	Zinc group	Boron group	Carbon group	Nitrogen group	Oxygen group	Fluorine group	Helium or Neon group
Period 1	H ^h																		He
Period 2	Li	Be												B	C	N	O	F	Ne
Period 3	Na	Mg												Al	Si	P	S	Cl	Ar
Period 4	K	Ca	Sc		Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Period 5	Rb	Sr	Y		Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Period 6	Cs	Ba	La	Ce–Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Period 7	Fr	Ra	Ac	Th–Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

^a Group 3 has scandium (Sc) and yttrium (Y). For the rest of the group, sources differ as either being (1) lutetium (Lu) and lawrencium (Lr), or (2) lanthanum (La) and actinium (Ac), or (3) the whole set of 15+15 **lanthanides** and **actinides**. IUPAC has initiated a project to standardize the definition as either (1) **Sc, Y, Lu and Lr**, or (2) **Sc, Y, La and Ac**.^[20]

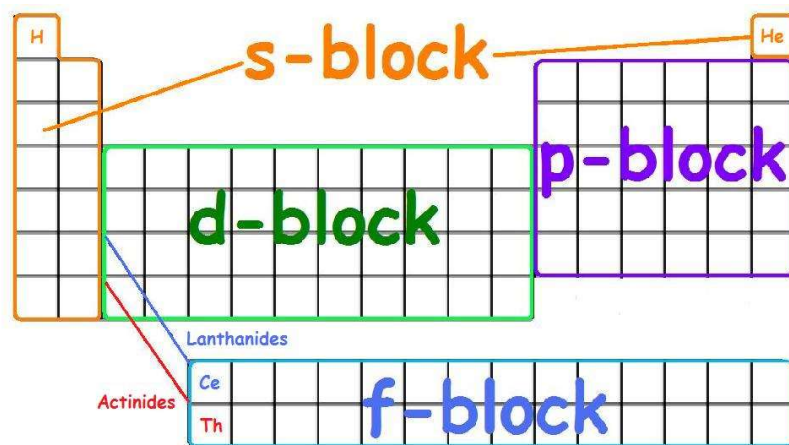
^b Group 18, the noble gases, were not discovered at the time of Mendeleev's original table. Later (1902), Mendeleev accepted the evidence for their existence, and they could be placed in a new "group 0", consistently and without breaking the periodic table principle.

^r Group name as recommended by IUPAC.

^h Hydrogen (H), while placed in group 1, is not considered to be part of the alkali metals.

Classification of the elements into BLOCKS

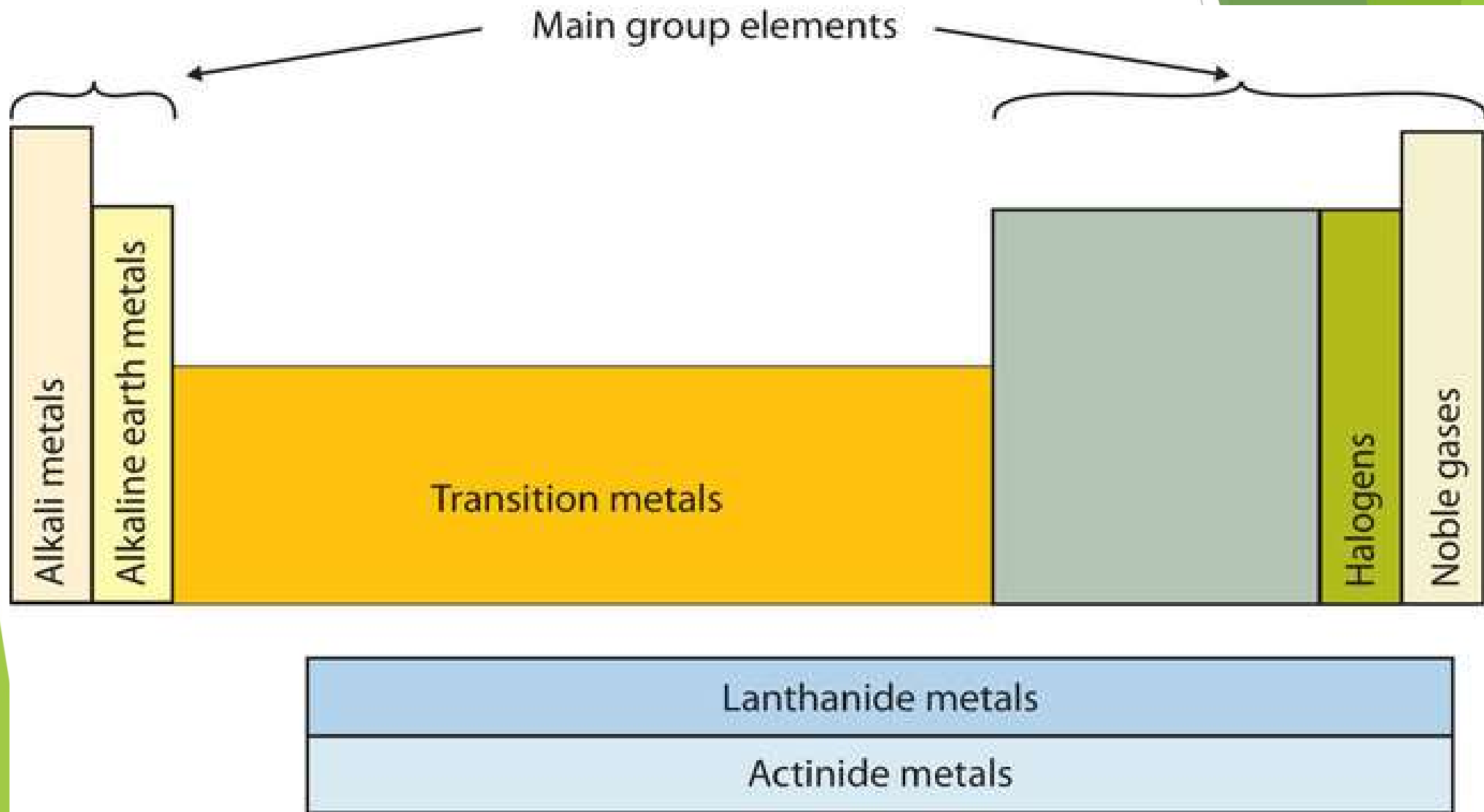
- ▶ The s-block elements have valence shell configuration s^1 or s^2 .
- ▶ The p-block elements have valence configuration s^2p^1 to s^2p^6 .
- ▶ The d-block elements have valence configurations in which d-subshells are being filled.
- ▶ The f-block elements have valence configurations in which f-subshells are being filled.



Classification of elements depending on electronic configuration:

- ▶ The Inert Gases (elements of the '0' group)
- ▶ The Representative Elements ('s' and 'p' block elements)
- ▶ The Transition Elements ('d' block elements)
- ▶ The Inner Transition Elements ('f' block elements)

Classification of elements depending on electronic configuration:



Classification of elements into metals, nonmetals & metalloids

- ▶ Elements are classified according to their properties into three major categories :
- ▶ (1) ***metals***,
- ▶ (2) ***nonmetals***, and
- ▶ (3) ***metalloids***.

What are Metals?

- ▶ Most elements are metals.
- ▶ they are divided into 3 groups:
 - ▶ (1) *alkali metals*,
 - ▶ (2) *alkaline earth metals*, and
 - ▶ (3) *transition metals*.
- ▶ The transition metals can be divided into smaller groups, such as the *lanthanides* and *actinides*.

Properties of Metals

- ▶ They are lustrous (shiny)
- ▶ malleable (can be hammered into sheets)
- ▶ ductile (can be drawn into wire)
- ▶ good conductors of heat and electricity.
- ▶ Except mercury, the metallic elements are solids at room temperature (about 20°C).
- ▶ These properties result from the ability to easily move the electrons in the outer shells of metal atoms.

Alkali Metals

- ▶ Group 1 is called alkali metals.
- ▶ Highly reactive and have +1 charge
- ▶ They are lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr).
- ▶ Alkali metals are common in nature and daily life.

Alkaline Earth Metals

- ▶ Group 2 is called alkaline earth metals.
- ▶ They have +2 charge.
- ▶ Smaller than those of the alkali metals.
- ▶ They are beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba) and radium (Ra).

Transition Metals

- ▶ Located in Groups 3 to 12.
- ▶ Iron and gold are examples of transition metals.
- ▶ Very hard, with high melting points and boiling points.
- ▶ Good electrical conductors and are very malleable.
- ▶ Generally show positive oxidation states of +2 and +3 and contain two incomplete energy levels ('s' & 'd' orbitals).

Periodic Table of the Elements

1 1IA 1A	2 IIA 2A											13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 H Hydrogen 1.008	2 He Helium 4.003											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3 Li Lithium 6.941	4 Be Beryllium 9.012											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
11 Na Sodium 22.99	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.799
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71 Lanthanide Series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon [222]
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103 Actinide Series	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [288]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]
		57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967	
		89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]	
		Alkali Metal	Alkaline Earth	Transition Metal	Basic Metal	Semimetal	Nonmetal	Halogen	Noble Gas	Lanthanide	Actinide						

Transition Metals...

- ▶ **Effective catalytic agents and generally form colored and complex compounds.**
- ▶ **The lanthanides (rare earth) and actinides (trans-uranium) are classes of transition elements.**

Metal Triads

- ▶ Transition metals with very similar properties are usually found together.
- ▶ The iron triad consists of iron, cobalt, and nickel.
- ▶ Below Iron triad- the palladium triad consists of ruthenium, rhodium, and palladium,
- ▶ And finally, the platinum triad consists of osmium, iridium, and platinum.

Lanthanides

- ▶ The lanthanides are silvery metals that tarnish easily.
- ▶ They are relatively soft metals, with high melting and boiling points.
- ▶ The lanthanides react to form many different compounds.
- ▶ These elements are used in lamps, magnets, lasers, and to improve the properties of other metals.

Actinides

- ▶ The actinides are in the row below the lanthanides.
- ▶ All of the actinides are radioactive, with positively charged ions.
- ▶ They are reactive metals that form compounds with most nonmetals.
- ▶ The actinides are used in medicines and nuclear devices.

Lanthanides & Actinides

- ▶ These elements have three incomplete outer levels ('s', 'd' & 'f' orbitals), and their properties are similar to transition elements.

Nonmetals

- ▶ A nonmetal is an element that does not exhibit the characteristics of a metal.
- ▶ Most of the nonmetals are gases (for example, chlorine and oxygen) or some are solids (phosphorus and sulfur).
- ▶ The solid nonmetals are usually hard, lacking metallic luster, and brittle substances.
- ▶ Bromine is the only liquid nonmetal.
- ▶ The halogens and the noble gases are two groups of nonmetals.

Metalloids or Semimetals

- ▶ **Metalloids, or semimetals** (B, Si, Ge, As, Sb, Te, Bi, Po), are elements having both metallic and nonmetallic properties.
- ▶ These elements, such as silicon (Si) and germanium (Ge), are usually good semiconductors—elements that, are poor conductors of electricity at room temperature and pure condition but become moderately good conductors at higher temperatures.

Halogens

- ▶ Halogens are in Group 17, form ions with a -1 charge. These are highly reactive. They are fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At).
- ▶ All the halogens react readily with metals to form salts, such as sodium chloride and calcium chloride (used in some areas as road salt).
- ▶ Compounds containing fluoride ions are added to toothpaste and the water supply to prevent dental cavities. Fluorine is also found in Teflon coatings on kitchen utensils.

Halogens...

- Chlorofluorocarbon (CFC) propellants and refrigerants are believed to lead to the depletion of Earth's ozone layer and contain both fluorine and chlorine.
- Bromine and iodine are less abundant than chlorine, and astatine is so radioactive that it exists in only negligible amounts in nature. These elements are also found in bleaches and disinfectants.

Noble Gases

- ▶ The noble gases are helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), and radon (Rn).
- ▶ The noble gases are composed of only single atoms, they are called monatomic.
- ▶ At room temperature and pressure, they are unreactive gases. Because of their lack of reactivity, once they were called inert gases or rare gases.
- ▶ However, the first chemical compounds containing the noble gases were prepared in 1962.

Noble Gases...

- Although the noble gases are relatively minor constituents of the atmosphere, natural gas contains substantial amounts of helium.
- Because of its low reactivity, argon is often used as an unreactive (inert) atmosphere for welding and in light bulbs. The red light emitted by neon in a gas discharge tube is used in neon lights.

Noble Gases...

The noble gases are not reactive.

Inactivity of the noble gases is due to

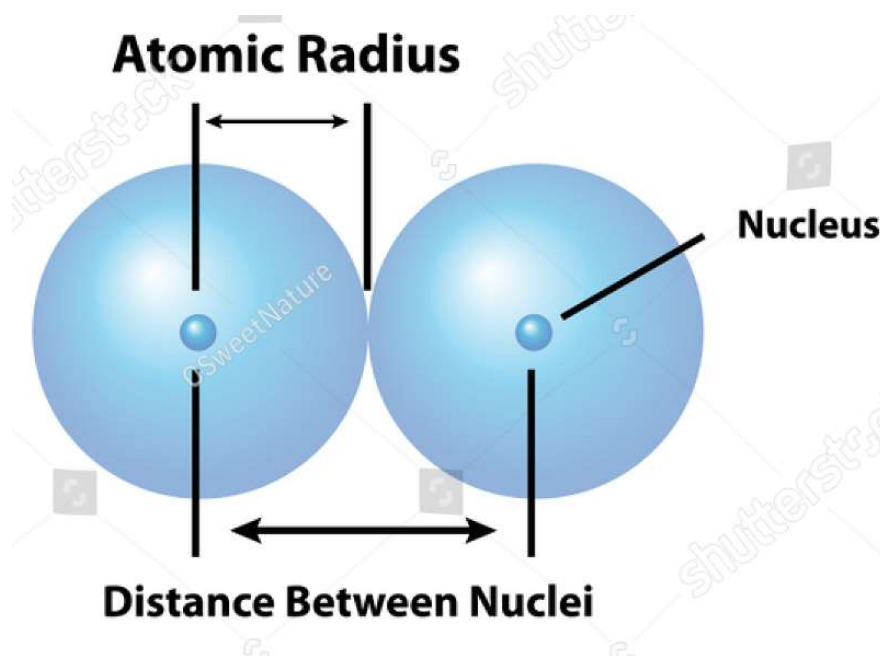
- (a) complete pairing of all electrons present
- (b) absence of any molecular orbital
- (c) stable energy state
- (d) very high ionization potential and
- (e) negligible electron affinities.

Periodic trends

Ionic Radii

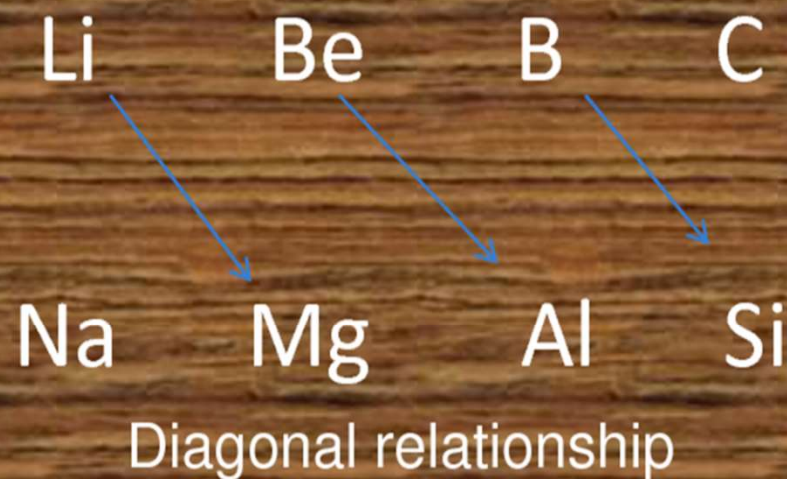
A monatomic ion, like an atom, is a nucleus surrounded by a distribution of electrons. The ionic radius is a measure of the size of the spherical region around the nucleus of an ion within which the electrons are most likely to be found.

If we imagine ions to be spheres of definite size, we can obtain their radii from known distances between nuclei in crystals



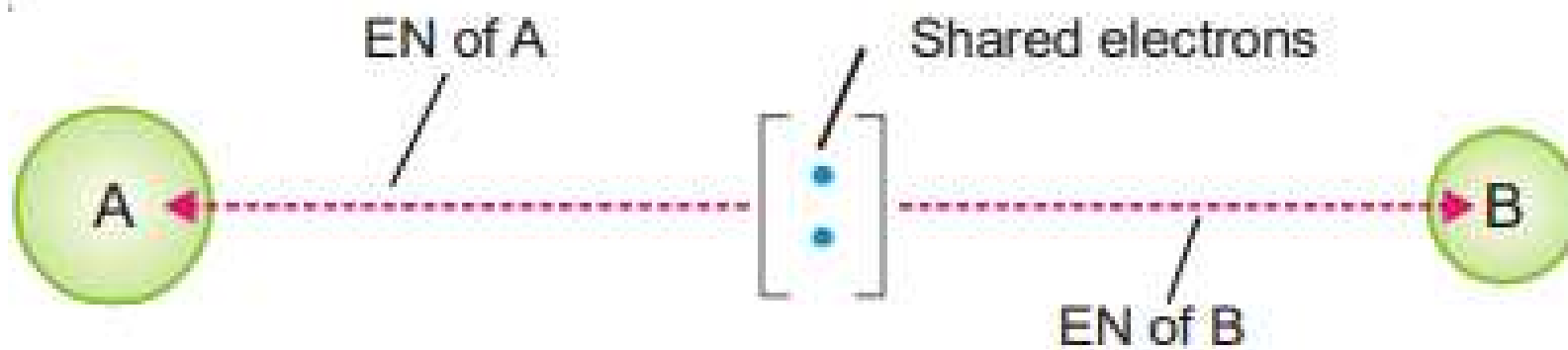
Diagonal Relationship

Some of the elements of second period show similarity in their properties with the elements of third period lying diagonally below it which is called diagonal relationship.



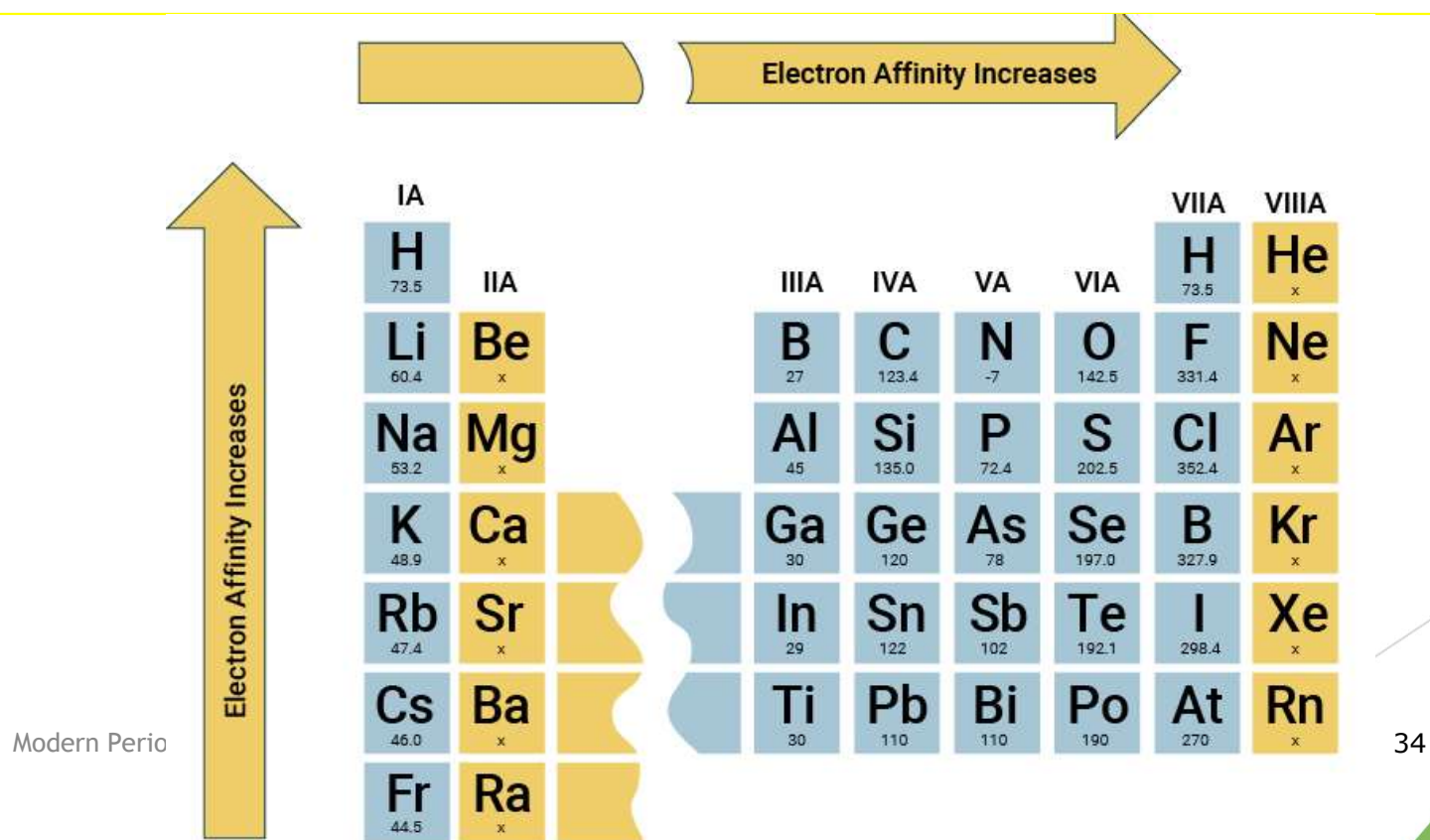
Some Definitions

- ▶ The **electronegativity** is a measure of the tendency of an atom to attract a bonding pair of electrons towards itself.



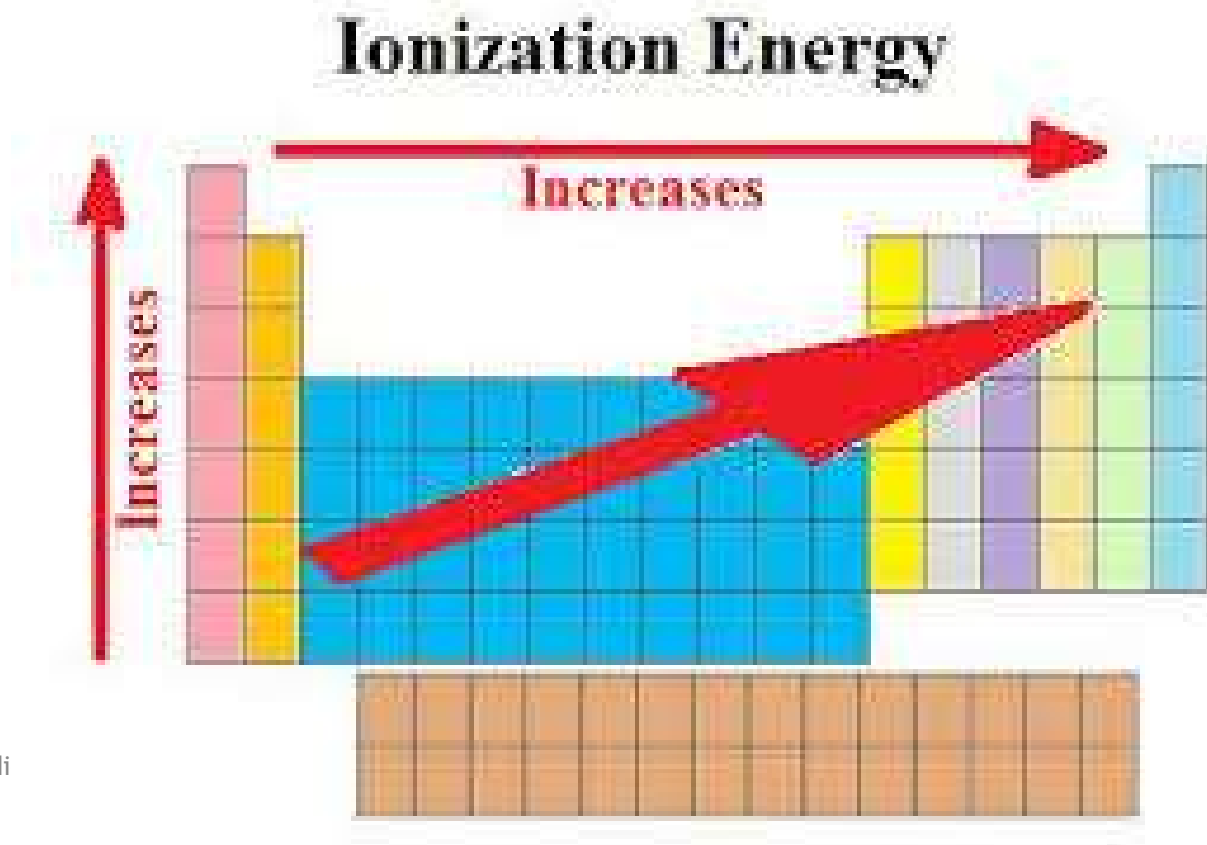
Some Definitions

- ▶ The **electron affinity** of an atom or molecule is defined as the amount of energy released or spent when an electron is added to a neutral atom or molecule in the gaseous state to form a negative ion ($X + e^- \rightarrow X^-$).

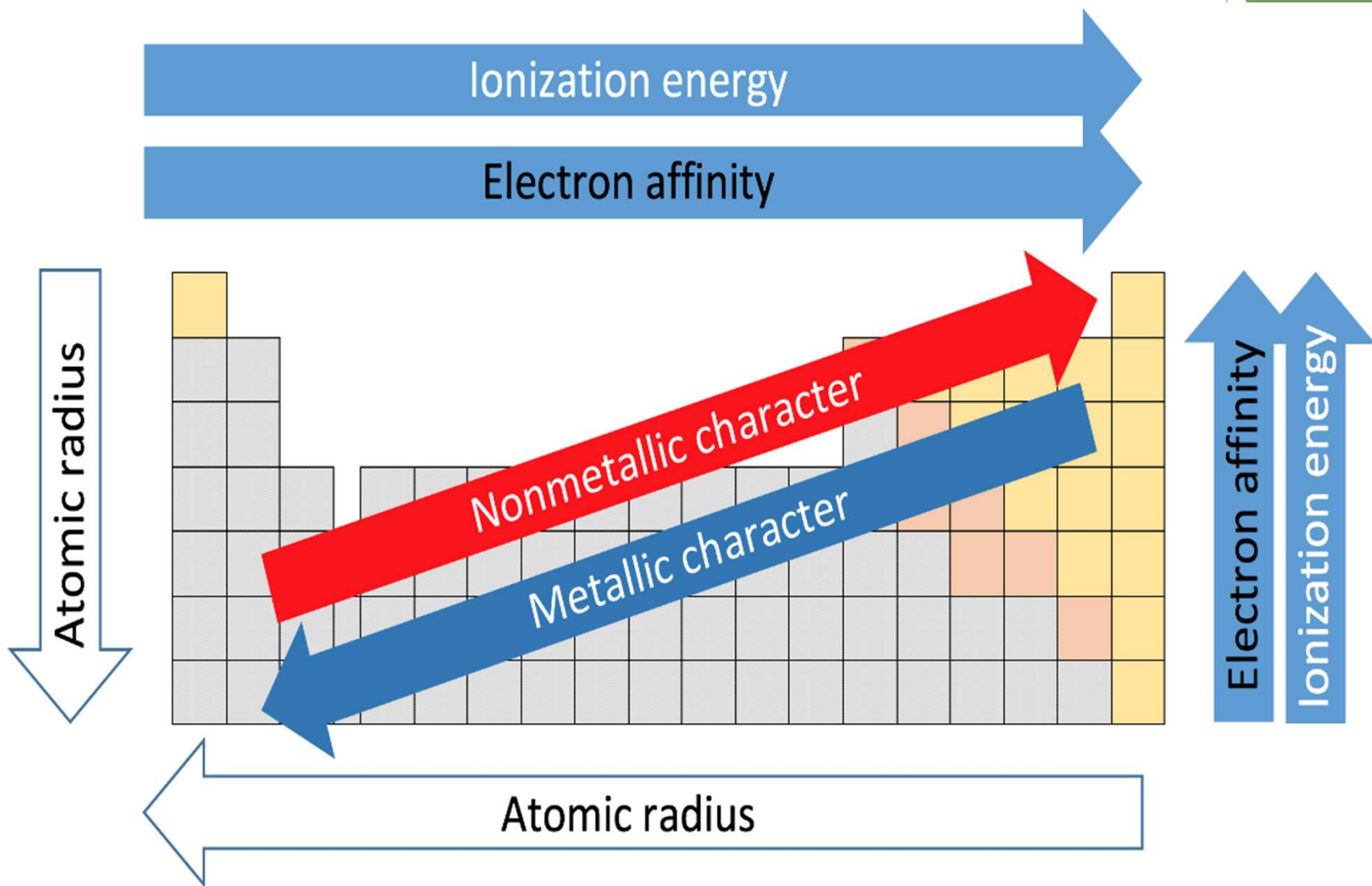


ionization energy

The **ionization energy** is the energy required to remove an electron from a gaseous atom or ion ($X + \text{energy} \rightarrow X^+ + e^-$).



Periodic trends



Periodic trends

- ▶ The periodic table helps predict some properties of the elements compared to each other. For example-
- ▶ Atomic size decreases as you move from left to right across the table and increases as you move down a column.
- ▶ Energy required to remove an electron from an atom increases as you move from left to right and decreases as you move down a column.
- ▶ The ability to form a chemical bond increases as you move from left to right and decreases as you move down a column.

Periodic trends...

- Generally the electron affinity value decreases with the increasing atomic radius because the electrostatic force of attraction decreases between the added electron and the atomic nucleus. This is due to an increase in the distance between them.

Naturally Occurring and Synthetic Elements

- ▶ Of the 118 known elements, 94 are found in appreciable quantities in nature. Between hydrogen (H, 1) and uranium (U, 92), only technetium (Tc, 43) and promethium (Pm, 61) are prepared artificially.
- ▶ All trans-uranium elements following uranium in the periodic table are synthetic.
- ▶ Elements are produced artificially in various nuclear transmutation reactions by neutrons or charged particles, including heavy ions.

Elements in the Human Body

- **The essential trace elements in the human body which make up about 0.1 % of body mass are iron (Fe), copper (Cu), zinc (Zn), iodine (I), and cobalt (Co).**
- **These elements are necessary for biological functions such as growth, transport of oxygen for metabolism, and defense against disease.**
- **Too much or too little of these elements in our body over an extended period of time can lead to serious illness, retardation, or even death.**



Next Class Chemical Bonding

Thank you